UNIVERSITY OF MINES AND TECHNOLOGY, TARKWA

FIRST SEMESTER EXAMINATION, APRIL 2024

COURSE NO: CE/EL/MC/MA 367
COURSE NAME: ARTIFICIAL INTELLIGENCE IN ENGINEERING
CLASS: CE/EL/MC/MA **TIME:** 3 HRS

Name**:** _____Index Number: _____

INSTRUCTION: Please circle the correct answer from the alternatives on the question paper.

1. Artificial Intelligence (AI) refers to……..
   a) Machines that can mimic human emotions
   b) Systems that can learn from data and adapt to new situations
   c) Machines with advanced physical capabilities
   d) The automation of repetitive tasks in manufacturing industries

2. A classic example of an AI-powered virtual assistant like Apple Siri or Nuance is …….
   a) Image recognition
   b) Language translation
   c) Speech recognition and natural language understanding
   d) Sentiment analysis

3. Which view or category of AI is closely associated with things which maximize goal achievement?
   a) Systems that think like humans
   b) Systems that think rationally
   c) Systems that act like humans
   d) Systems that act rationally

4. According to the Turing Test, when can a machine be considered to exhibit human-like intelligence?
   a) When it can solve complex mathematical problems
   b) When it can pass for a human in a conversation
   c) When it can outperform humans in cognitive tasks
   d) When it can perform tasks faster than humans

5. What does it mean for an agent to be rational?
   a) The agent always makes perfect decisions without any errors.
   b) The agent follows a set of predefined rules without deviation.
   c) The agent selects actions that maximize its expected utility given its knowledge and goals.
   d) The agent acts randomly without any consideration for its goals or environment.

6. What do actuators represent in the PEAS framework?
   a) Components that perceive the environment
   b) Components that determine the agent's performance measure
   c) Components that execute actions in the environment
   d) Components that monitor changes in the environment

7. In supervised learning, what does a labeled dataset consist of?
   a) Only input features.
   b) Only output labels.
   c) Only restricted labels.
   d) Input features along with corresponding output labels.

8. Which search algorithm always selects the deepest unexpanded node in the search tree?
   a) Breadth-First Search (BFS)
   b) Depth-First Search (DFS)
   c) Uniform Cost Search (UCS)
   d) A* Search

9. The following are fundamental component of a search problem except ……..
   a) Initial state
   b) Goal state
   c) Transition model
   d) Learning algorithm

10. What does the acronym "IF-THEN" represent in production rules?
    a) Input-Filter-Test-Hypothesize-Notify
    b) Identify-Filter-Test-Hypothesize-Notify
    c) If-This-Then-That
    d) If-Filter-Test-Then-Hypothesize-Notify

11. In an environment with partial observability, what challenge does the agent face?
    a) It cannot perceive the entire state of the environment
    b) It lacks the ability to execute actions
    c) It has limited memory capacity
    d) It cannot accurately measure the utility of different states

12. How does a rational agent differ from a reflex agent?
    a) A reflex agent always selects actions randomly, while a rational agent selects actions based on its goals.
    b) A reflex agent follows a fixed set of rules, while a rational agent adapts its behavior based on feedback.
    c) A reflex agent ignores its environment, while a rational agent considers its goals and selects actions to achieve them.
    d) A reflex agent selects actions based on the current percept only, while a rational agent considers future consequences.

13. What are association rules in unsupervised learning?
    a) Rules that predict a target variable based on input features.
    b) Rules that describe the relationship between two variables.
    c) Rules that identify frequent itemsets in transaction data.
    d) Rules that cluster similar data points together.

14. Identify the algorithm best used to perform classification task?
    a) Linear Regression
    b) K-Means Clustering
    c) Decision Trees
    d) Principal Component Analysis (PCA)

15. Which of the following describes properties of Search Algorithm?
    I. Complete
    II. Optimality
    III. Time Complexity
    IV. Space Complexity

a) I and II
b) II and III
c) I, II and III
d) I, II, III and IV

16. An example of a performance measure for a vacuum cleaner agent is …….
a) The number of rooms cleaned per minute
b) The type of flooring in the environment
c) The size of the dust bin in the vacuum cleaner
d) The color of the walls in the environment

17. …… is a type of environment is characterized by the fact that the next state of the environment is completely determined by the current state and the action taken by the agent?
a) Deterministic environment
b) Stochastic environment
c) Static environment
d) Dynamic environment

18. Which uninformed search algorithm guarantees the optimal solution if the step costs are non-negative?
a) Depth-First Search (DFS)
b) Breadth-First Search (BFS)
c) Uniform Cost Search (UCS)
d) Iterative Deepening Depth-First Search (IDDFS

19. The technique that uses mathematical equations to represent uncertain or imprecise information is..
a) Semantic networks
b) Production rules
c) Predicate logic
d) Fuzzy logic

20. Which search algorithm combines the advantages of both Breadth-First Search (BFS) and Depth-First Search (DFS)?
a) Uniform Cost Search (UCS)
b) A* Search
c) Iterative Deepening Depth-First Search (IDDFS)
d) Bidirectional Search

21. What is the purpose of the alpha value in the alpha-beta pruning algorithm?
a) To represent the maximum score encountered along the search path
b) To represent the minimum score encountered along the search path
c) To prune branches of the search tree
d) To represent the worst-case scenario for the player

22. What is the main advantage of using alpha-beta pruning over the minimax algorithm?
a) It guarantees finding the optimal solution
b) It reduces the time complexity of the search
c) It allows for deeper searches in the game tree
d) It can handle games with chance elements

23. Which algorithm is useful for modelling environments where adversary agents are based on chance?
a) Minimax algorithm
b) Alpha beta pruning algorithm

c) Expectimax algorithm
d) Maximin algorithm

24. Which AI technique is commonly used in recommendation systems such as those used by Amazon or Netflix?
a) Natural Language Processing
b) Reinforcement Learning
c) Deep Learning
d) Collaborative Filtering

25. The best data structure to implement Breadth-First Search (BFS) is ……. while Depth-First Search (DFS) is ……..
I. LIFO Stack and FIFO Queue
II. FIFO Stack and LIFO Queue
III. LIFO Stack and FIFO Priority Queue

a) I
b) II
c) III
d) None of the above

26. What does the term overfitting refer to in machine learning?
a) The model learns patterns from the training data that do not generalize well to unseen data
b) The model fails to capture complex patterns in the training data
c) The model performs well on the training data but poorly on the testing data
d) The model performs equally well on both the training and testing data

27. If Player 1 is Max and Player 2 is Min, determine the utility value of player 1 in figure 1?

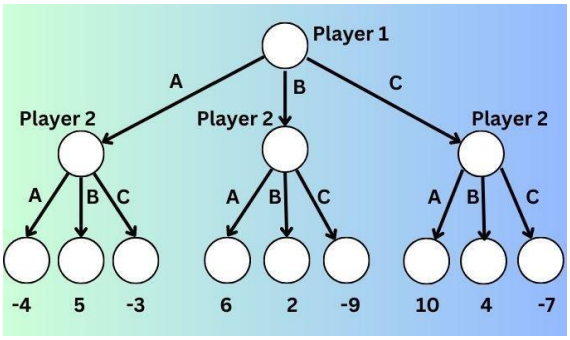

Figure 1

a) 10
b) -9
c) -7
d) -4

28. What is the time complexity of minimax algorithm in figure 1?
a) $O(b^m)$
b) O(bm)
c) $O(b^{m/2})$
d) O(bm/2)

29. Which of the following is an example of an actuator for a robot agent?

a) Camera
b) Wheel motors
c) Microphone
d) Temperature sensor

30. The reasoning technique best applied by Factory Floor Scheduling is ……...
a) Horizontal Planning
b) Genetic Algorithm
c) Pattern Matching
d) Image Processing

31. What problem does the horizon effect pose for search algorithms?
a) It may lead to suboptimal solutions due to limited lookahead.
b) It may result in an infinite loop during the search process.
c) It may cause the search algorithm to overlook important features of the environment.
d) It may increase the computational complexity of the search algorithm.

32. When constructing a decision tree, which attribute is chosen at each node to split the data based on impurity measures?
a) The one with the highest impurity
b) The one with the lowest impurity
c) The one with the highest information gain or lowest impurity
d) The one with the highest variance

33. Which function is used in the modeling and solving of games in AI?
a) Search
b) Logics
c) Reasoning
d) Heuristic

34. In adversarial games, what does the term "utility function" represent?
a) The probability of winning the game
b) The total number of moves available to a player
c) The score or payoff associated with a particular game state
d) The number of players involved in the game

35. Which statement(s) represent problem-solving process in AI?
I. Formulating the problem
II. Analyzing the problem
III. Choose the solution
IV. Finding solutions

a) I, II, and IV only
b) I, III, and IV only
c) I and IV only
d) I, II, III, and IV only

36. How does the concept of minimax apply to adversarial games?
a) It aims to minimize the number of moves required to win.
b) It aims to maximize the number of moves available to the opponent.
c) It aims to minimize the maximum possible loss for a worst-case scenario.
d) It aims to maximize the total number of moves made by both players.

37. In a dynamic environment, what characteristic distinguishes it from a static environment?
   a) The environment changes while the agent is deliberating
   b) The agent has complete knowledge of the environment
   c) The environment is partially observable
   d) The outcomes of actions are determined by chance

38. If a multi-agent system has n agents, each of which has m possible moves, the search space increases to how many possible states?
   a) $mn\text{-}^{mn}$
   b) m^n
   c) mn-mn
   d) m^n-mn

39. Tic tac toe game is a classic example of …
   a) Deterministic games
   b) Nondeterministic games
   c) Zero Sum Games
   d) Perfect Games

40. Which of the following English sentences is a faithful paraphrase of the following formula?

$$HappyParent = Parent \wedge \exists \text{hasChild } (\text{Doctor} \wedge \forall \text{hasChildDoctor})$$

   a) Happy parents are precisely those parents all of whose children are either doctors or have a child who is a doctor
   b) Parents of doctors who also have children who have a doctor as a child, are happy parents
   c) All happy parents have a child that is a doctor, or the parent of a doctor
   d) Happy parents are precisely those parents who have a child that is a doctors and this doctor's children are doctors as well

SECTION B

Please answer question ONE and any other question in this section in the provided ANSWER BOOKLET. Each question carries 10 marks.

Question One

a. With aid of diagram, describe the concept of these agent types and how they make decisions.
   i. Goal Based Agents
   ii. Learning Agents [4 marks]

b. Using the PEAS describe the task environment of
   i. Taxi Driving Agent
   ii. Typing Tutor Agent
   [2 marks]

c. Describe the environmental properties of Medical Diagnosis Expert System [2 marks]

d. Briefly explain how are AI-driven platforms are enhancing students' learning experiences within your educational program? [2 marks]

Question Two

a. Differentiate between the following.
   i Perfect Information and Imperfect Information
   ii. Deterministic and Non-deterministic games
   [4 marks]

b. Describe the process of constructing a game tree for a tic-tac-toe. Include the initial state, possible moves, and terminal states in your explanation. [3 marks]

c. Consider a decision node in a game tree with three possible actions: A, B, and C. The probability of each action leading to different outcomes is as follows:
   Action A: 0.36 (Outcome 1), 0.5 (Outcome 2), 0.2 (Outcome 3)
   Action B: 0.4 (Outcome 1), 0.4 (Outcome 2), 0.2 (Outcome 3)
   Action C: 0.2 (Outcome 1), 0.3 (Outcome 2), 0.5 (Outcome 3)
   Calculate the expected imax value for each action assuming the utility values for outcomes are 3, 5, and 2 respectively.
   [3 marks]

Question Three

a. Discuss the role of knowledge representation in AI agents and provide two examples of different representation methods. [2 marks]

b. Consider the training examples below for a binary classification problem.

Customer ID	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1

a) Compute the Gini index for the overall collection of training examples.
b) Compute the Gini index for the Customer ID attribute.
c) Compute the Gini index for the Gender attribute.
d) Compute the Gini index for the Car Type attribute using multiway split.
e) Compute the Gini index for the Shirt Size attribute using multiway split.
f) Which attribute is better, Gender, Car Type, or Shirt Size?
g) Explain why Customer ID should not be used as the attribute test condition even though it has the lowest Gini. [7 marks]

Examiner: Ezekiel Mensah Martey (PhD)

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